An Ultrawideband 1-Bit Reconfigurable Reflectarray Element With Improved Angular Stability

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***Abstract*—In this letter, a 1-bit ultrawideband reconfigurable reflectarray (RRA) element with improved angular stability is proposed. Based on the magneto-electric (ME) dipole, a novel central-patch probe structure is introduced, wherein two diodes are connected to opposite sides of the probe to control the two resonant modes. This configuration yields a fully symmetric RRA element, which balances and stabilizes the current distributions associated with both resonant modes, thereby enabling an ultrawideband 1-bit phase response. Furthermore, rows of metallic vias are incorporated on both sides of the element to enhance the angular stability of the 1-bit phase bandwidth. Consequently, the proposed RRA element achieves a 1-bit relative phase bandwidth of 58% (8.97 GHz to 16.44 GHz) and maintains this bandwidth at 33% for oblique incidence angles up to 50°. To experimentally validate the design, a 1 × 2 elements prototype was fabricated and characterized using the rectangular waveguide method. The measured results exhibit good agreement with the simulations, thereby verifying the ultra-wideband performance of the proposed element. System-level simulations of a 16 × 16 array confirm that the proposed element significantly enhances the gain bandwidth and wide-angle scanning performances. These characteristics make the proposed RRA element a strong candidate for future radar, satellite communication, and reconfigurable intelligent surface applications.**

***Index Terms*—Magnetoelectric dipole, metasurface, reconfigurable intelligent surface, reconfigurable reflectarray, wideband.**

I. Introduction

With the continuous advancement of wireless communication technologies and the widespread deployment of radar detection systems, antennas for electromagnetic wave transmission and reception are facing increasingly stringent performance requirements, including beam-steering capability, wide bandwidth, and high gain. In recent years, reconfigurable reflectarrays (RRAs) have drawn significant attention owing to their flexible beam control capacity, compact size, lightweight structure, low profile, cost-effectiveness, and ease of fabrication [1], [2], [3], [4], [5], [6], [7]. Specifically, the 1-bit RRA—a type of reflectarray that employs only two discrete phase states (typically 0° and 180°) to manipulate the reflection phase—is characterized by its structural simplicity, low cost, and ease of design. Providing considerable flexibility in beamforming, it demonstrates significant promise for next-generation communication systems [8], [9], [10], [11].

Broadband capability has emerged as a pivotal trend in modern wireless communications, as broadband antennas enable efficient operation over extended frequency ranges, thereby enhancing system performance and reliability [12]. The 1-bit phase relative bandwidth of an RRA element—defined as the fractional bandwidth over which the phase difference is maintained within a specific tolerance of 180° (e.g., 180° ± 20°)—serves as a critical determinant of the RRA's overall broadband performance [13]. However, the majority of existing 1-bit RRA elements are predominantly based on tunable resonant structures [14], which are inherently narrowband and thus restrict further bandwidth enhancement. To address this challenge, [15] proposed a simplified 1-bit broadband RRA element utilizing a single p-i-n diode; however, it yielded a 1-bit relative bandwidth of merely 12%, which remains insufficient for many broadband applications. In [16], a multilayer stacked configuration was employed to extend the phase bandwidth within the Ku band, albeit at the cost of a higher profile and greater fabrication complexity. Similarly, a cost-effective design based on an ME dipole structure was reported in [17]; nevertheless, its asymmetric geometry induces a nonuniform surface current distribution, thereby constraining the achievable relative bandwidth.

In this work, a novel central-patch probe located at the center of the ME dipole is proposed to achieve a fully symmetric scattering structure. Two diodes are integrated between the probe and the electric dipole patches to control the two resonant modes. Consequently, an ultrawideband 1-bit phase response spanning 8.97GHz–16.44 GHz is achieved. In addition, angular-stabilizing structures are introduced to preserve the 1-bit bandwidth performance under oblique incidence. To comprehensively evaluate the proposed design, experimental validation was conducted using a 1 × 2 prototype characterized via the rectangular waveguide (RWG) method [18], alongside system-level validation based on a 16 × 16 array simulation. The results demonstrate the ultra-wideband characteristics and angular stability of the element, as well as its contribution to enhancing array performance. To the best of the authors' knowledge, this design achieves the widest 1-bit phase bandwidth among reported X- and Ku-band 1-bit RRA elements, demonstrating superior performance even under large angles of incidence.

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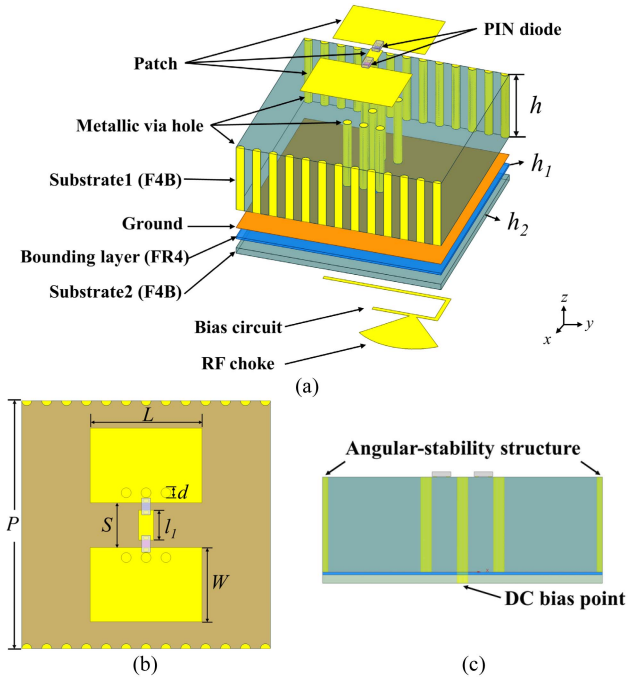


Fig. 1. Structure of the proposed ultrawideband 1-bit RRA element. (a) Exploded view. (b) Top view. (c) Side view.

TABLE I
DIMENSIONS OF THE ULTRAWIDEBAND 1-BIT RRA ELEMENT (UNITS: MM)

Parameters	P	L	W	S	l_1
Values(mm)	10	4.5	3	1.8	1.2
Parameters	d	h	h_1	h_2	
Values (mm)	0.4	3.4	0.1	0.3	

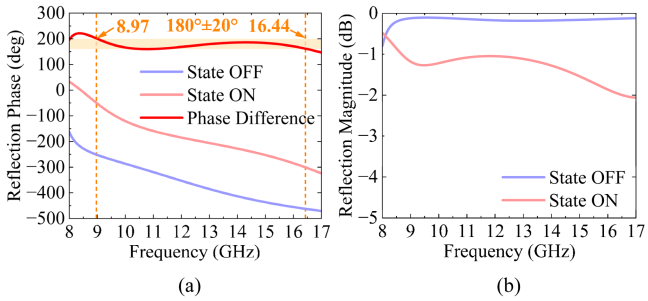


Fig. 2. Simulated reflection (a) phases and (b) magnitudes of the proposed ultrawideband RRA element under the normal x-polarized incident wave with different state.

II. ULTRAWIDEBAND AND ANGULARLY STABLE RRA ELEMENT

A. Element Configuration

Fig. 1 depicts the detailed structure of the proposed ultrawideband RRA element. The structure is x-polarized and consists of two dielectric substrates (F4B, $\epsilon_r = 2.2$, $\tan\delta = 0.007$). A 0.1 mm-thick bonding layer (FR4, $\epsilon_r = 4.2$, $\tan\delta = 0.02$) is placed between the substrates to facilitate fabrication. The upper layer of Substrate1 comprises two large metallic patches and a novel central-patch probe. The two large patches serve as an electric dipole, with two symmetrically placed p-i-n diodes bridging them. Six metallic vias are embedded in Substrate1, forming a loop-like structure in conjunction with the metal ground plane between them. This configuration acts as an equivalent magnetic dipole. The central-patch probe, along with the vias beneath it, is used for biasing the diodes. A bias voltage is applied from the dc bias point located on the bottom layer of Substrate2 to simultaneously switch the two p-i-n diodes between ON and OFF states, thereby realizing two reconfigurable reflection states of the RRA element. It is worth noting that the central-patch probe renders the proposed element fully symmetric along both the x- and y-axes, resulting in a more uniform and stable current distribution for both resonance states, thereby expanding the operating bandwidth. To prevent high-frequency signals from leaking into the dc path, a sector-shaped stub is incorporated into the bottom biasing circuit to serve as an RF choke. Furthermore, two rows of metallic vias are placed around the element, adjacent to the electric dipole patches, to enhance angular stability.

A p-i-n diode, model MACOM MA4AGFCP910, is used to control the reflection states of the proposed RRA element in this design. For ON or OFF state, the p-i-n diode is modeled as a series of lumped resistance (R) and inductance (L) or capacitance (C) and inductance (L), respectively [8]. More detailed specifications of this component can be found in the manufacturer's datasheet [19]. By applying distinct dc bias voltages to the bottom bias point, the two top-layer p-i-n diodes are simultaneously switched between the ON and OFF states, thereby exciting two reconfigurable resonant modes of the ME dipole. Through the optimization of the element's critical geometric parameters, the two modes can be tuned to coincide at the same resonant frequency with comparable excitation magnitudes. This alignment ensures stable 1-bit reflection characteristics. The optimized geometric parameters are summarized in Table I.

B. Ultrawideband Characteristic and Operating Mechanism

The reflection performances in terms of phase and magnitude under normal incidence are shown in Fig. 2(a). The phase difference between the two switching states of the RRA element remains within $180° \pm 20°$ across the 8.97GHz–16.44 GHz frequency range, achieving a 1-bit phase bandwidth of 58%, which demonstrates ultrawideband performance. Fig. 2(b) illustrates the amplitude responses of the RRA element in both switching states. It is evident that the element exhibits a wide resonant bandwidth in both states, consistent with the broadband radiation characteristics of ME dipoles [20]. In the ON state, the element exhibits higher loss, primarily due to the inevitable energy dissipation caused by resistive heating as the current flows through the p-i-n diodes. Nevertheless, the reflection magnitudes in both states are larger than -2 dB, which is quite acceptable for 1-bit RRA design.

To further validate the operating principle of the proposed RRA element, current distribution analyses were performed and discussed. Fig. 3 shows the current distribution of the RRA element under x-polarized wave incidence. Solid black lines indicate the directions of the current vectors. It is observed that the proposed RRA element operates in a magnetic-dipole resonance mode when the diodes are reverse-biased (OFF state) and in an electric-dipole resonance mode when forward-biased (ON state) [17]. It should be noted that the resonance mode associated with each reconfigurable state remains stable across

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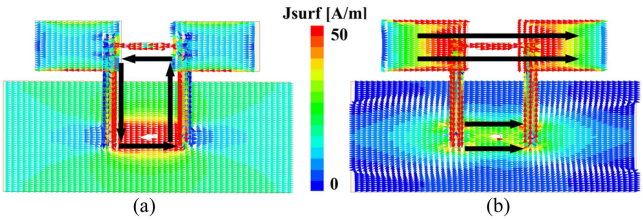


Fig. 3. Current distribution of the proposed RRA element at 12.5 GHz: (a) Magnetic dipole resonance for OFF state. (b) Electric dipole resonance for ON state.

the entire operating band. This indicates that the ME dipole modes excited by the proposed element exhibit ultrawideband characteristics. This performance is attributed to the proposed central-patch probe structure, which enables the RRA element to exhibit a fully symmetric scattering configuration and maintain a stable current distribution for both resonance states across this ultra wide bandwidth.

C. Improved Angular Stability Characteristics

In practical RRA applications, the oblique incidence of electromagnetic waves is inevitable. Compared to normal incidence, the performance of an RRA typically deteriorates under oblique incidence. This degradation is commonly attributed to multiple-reflection interference and the interactions between adjacent RRA elements [21]. Given that the proposed element utilizes a structure comprising two dielectric substrates separated by a metallic ground plane, this analysis primarily focuses on the mutual coupling between adjacent elements.

To mitigate this effect, two rows of metallic vias are introduced along both sides of the element in the E-plane. The spacing between these vias is significantly smaller than the operating wavelength, allowing them to function as an electric wall. This configuration introduces an additional boundary condition that effectively suppresses mutual coupling, thereby minimizing performance degradation under oblique incidence.

Fig. 4 compares the simulated reflection characteristics with and without metallic vias under different incident angles (θ_i). In the absence of metallic vias, the reflection characteristics exhibit significant variations in both amplitude and phase as θ_i increases. Specifically, when θ_i exceeds 20°, anomalous resonances—manifested as significant fluctuations in the phase curve and sharp dips in the amplitude curve—occur around 11 GHz, disrupting the continuity of the phase difference curve and resulting in a substantial reduction in the 1-bit phase bandwidth. This phenomenon is primarily attributed to undesired mutual coupling between adjacent elements, which severely degrades the reflection performance of the RRA and leads to limited angular stability. With the introduction of metallic vias, these anomalous resonances are effectively eliminated, and the reflection phase difference becomes significantly more stable. Notably, the element maintains a remarkable relative bandwidth of 33% (10.05–14.11 GHz), even under oblique incidence angles up to 50°. This result demonstrates that the incorporation of metallic vias significantly enhances the performance of the RRA element under oblique incidence.

D. Performance Evaluation of RRA System

To comprehensively evaluate the performance of the proposed RRA element, a full RRA system comprising 16 × 16 elements

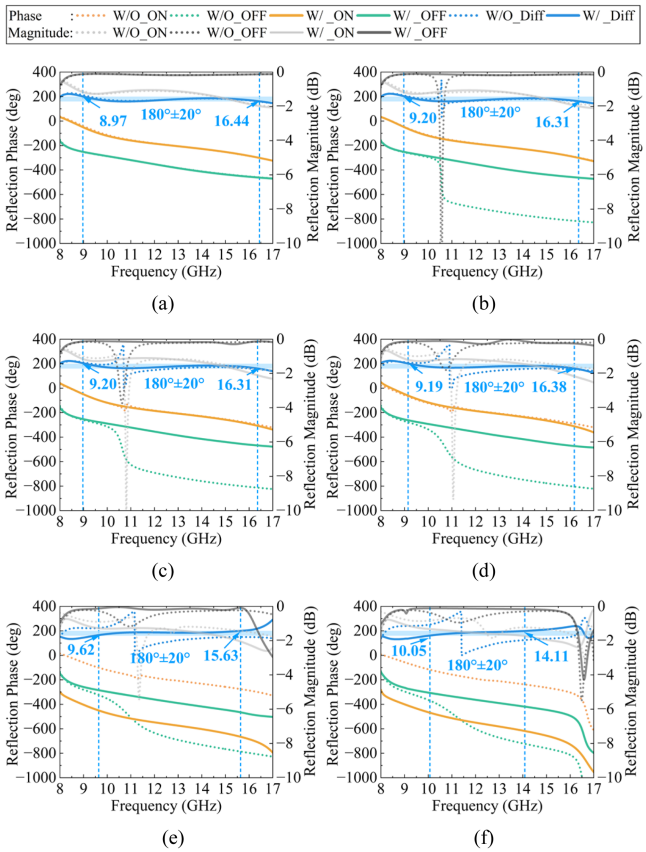


Fig. 4. Simulated phase responses of the RRA element with and without metallic shorting vias under different incident angles: (a) 0°. (b) 10°. (c) 20°. (d) 30°. (e) 40°. (f) 50°.

and illuminated by a wideband horn antenna was analyzed using Ansys HFSS. The feed horn is offset by 20° from the z-axis to mitigate feed blockage. The focal length-to-diameter ratio (F/D) is optimized to 1.43 to maximize the aperture efficiency.

The simulated radiation performance, including radiation patterns, gain, and aperture efficiency, is illustrated in Fig. 5. Over the operating frequency band of 8.97 Ghz-16.44 GHz, the simulated 1 dB and 3 dB gain bandwidths in the E-plane are 28.6% and 46.2%, respectively, demonstrating superior performance compared to state-of-the-art designs. Furthermore, a peak aperture efficiency of 31.4% is achieved within this band, yielding a realized gain of 22.1 dBi at 12 GHz. In terms of beam-scanning performance, the main beam is steered from 0° to 60° in 10° steps. Well-defined pencil beams are achieved in all desired directions, exhibiting low scan loss and the absence of grating lobes. Furthermore, the proposed RRA maintains wide-angle scanning capabilities across the entire operating bandwidth. These results demonstrate that the proposed ultrawideband element significantly enhances the bandwidth performance of the RRA system while maintaining excellent angular stability.

III. Experimental Validation With Rectangular Waveguide Measurement

To experimentally validate the proposed RRA element, a 1 × 2 element prototype was fabricated and characterized using the RWG setup, as shown in Fig. 6(a). Fig. 6(b) presents a photograph of the waveguide assembly with the prototype element, where the dc bias for the diodes is supplied via soldered wires.

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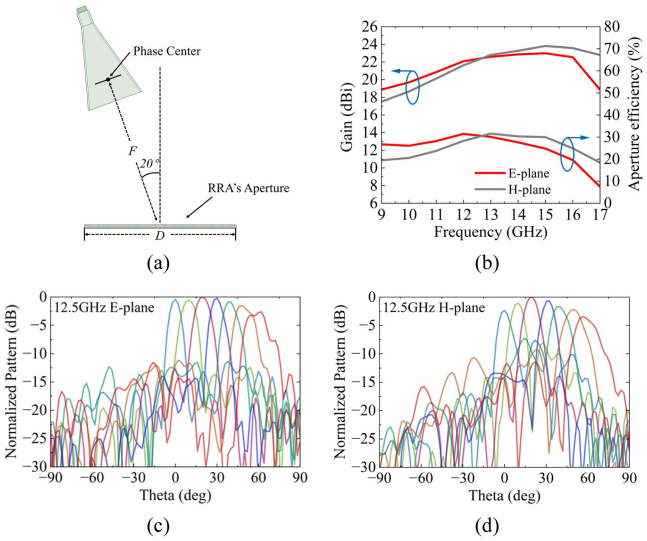


Fig. 5. Simulated performances of the analytical RRA system. (a) System configuration. (b) Gain and aperture efficiency versus frequency for the E- and H-planes. (c) Scanned radiation patterns from 0° to 60° in the E-plane at 12.5 GHz. (d) Scanned radiation patterns in the H-plane at 12.5 GHz.

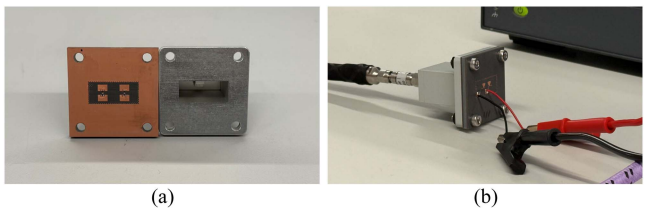


Fig. 6. Photos of fabricated prototype and experiment. (a) Fabricated 1 × 2 RRA element prototype. (b) Rectangular waveguide measurement setup.

During the measurement, the prototype element was mounted on the flange of the waveguide, which was connected to a vector network analyzer (Keysight E5063A) at the input port. Following the TRL calibration, the reflection coefficient at the surface of the prototype was directly measured. It should be noted that although the RWG method inherently characterizes reflection under oblique incident waves with fixed polarization—differing from ideal normal incidence—it provides a fast and cost-effective method for evaluating RRA performance and has been widely adopted in large-scale reflectarray research [9], [22], [23].

To accommodate the ultrawide operating bandwidth of the proposed RRA element, standard WR-90 and WR-75 waveguides were utilized to cover the 8.2 GHz to–12.4 GHz and 10 GHz to 15 GHz bands, respectively. Corresponding full-wave simulations modeling the waveguide setup were conducted for validation. Fig. 7 presents a comparison between the simulated and measured results, demonstrating good agreement in both reflection phase and magnitude; this consistency validates the ultrawideband performance of the proposed 1-bit RRA element. Minor discrepancies, such as additional spurious resonances observed in the measurements, are primarily attributed to fabrication tolerances and misalignment between the waveguide and the prototype during assembly.

To provide a comprehensive evaluation of the proposed design, Table II benchmarks its key metrics against those of

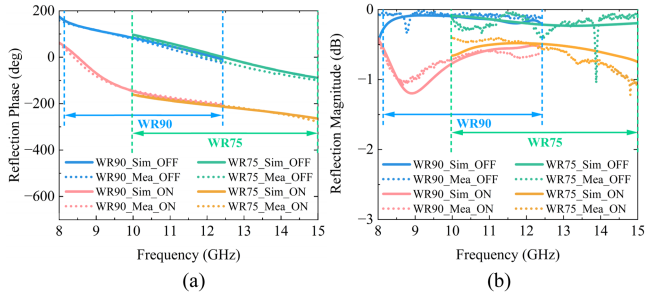


Fig. 7. Measured reflection performance of the proposed ultrawideband RRA element prototype. (a) Phases. (b) Magnitudes.

TABLE II
COMPARISONS BETWEEN THE PROPOSED AND REPORTED 1-BIT RRA

Ref. No.	Freq. (GHz)	1-bit Phase Bandwidth*	Angular-Stabilizing Structure	Gain BW (%)	Scan Loss# (dB)
[13]	10.5-17.5	38%	No	1-dB: 25 1.5-dB: 32	N/A
[15]	4.7-5.3	12%	No	1-dB: 8.4	3.1 @ 50°
[16]	12.9-16.5	24%	No	1-dB: 22.5	4.7 @ 60°
[17]	10-15	40%	No	1-dB: 22.3 3-dB:38.4	5.3 @ 60°
[23]	6.75-11.25	50%	No	N/A	N/A
This Work	**8.97-16.44**	**58%**	**Yes**	**1-dB: 28.6 (Sim.) 3-dB: 46.2 (Sim.)**	**3.6 @ 60° (Sim.)**

*Criteria: 180°±20°. #Gain Loss at the maximum scan angle (H-plane).

state-of-the-art wideband 1-bit RRA. As observed, the proposed element achieves an ultrawide 1-bit phase bandwidth of 58% while simultaneously featuring angular stability—a capability that has not been addressed in existing works. Moreover, the system-level performance in terms of gain bandwidth and scan loss demonstrates the superiority and potential of the proposed design. To the best of the authors' knowledge, the proposed design achieves the widest 1-bit phase bandwidth among reported X- and Ku-band RRA elements, while demonstrating superior angular stability.

IV. CONCLUSION

This letter presents a novel 1-bit ultrawideband RRA element with enhanced angular stability. The element achieves a 58% 1-bit relative phase bandwidth (8.97 GHz to 16.44 GHz) and retains a 33% bandwidth under oblique incidence up to 50°. To validate the design, a 1 × 2-element prototype was fabricated and characterized using the RWG method. The measured results exhibit good agreement with simulations, thereby verifying the excellent 1-bit phase bandwidth and the associated low loss. Furthermore, system-level performance was evaluated through numerical simulations of a 16 × 16 array. The results confirm that the proposed element significantly enhances the gain bandwidth and wide-angle scanning performance of the RRA system. These characteristics make the proposed element a strong candidate for wideband applications in next-generation wireless systems.

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